Persona Consistency in LLMs - Game Non person characters(NPCs)

Why I selected this project

Games nowadays, are starting to use AI language models for NPC talking instead of writing fixed scripts for every single character every time is what is happening now. But there is a problem, this model, sometime it can forget what before they said it, in long talking is happening this, or contradicting own self they are, or sometimes just break character and start talking like a normal chatbot, which for game feeling this is very bad thing. Idea for do this project is what this give me, want I to checking, does really happen this, when NPC persona we making and talk with it long time we do, and also if fixing it we can with easy method some, or maybe is not possible at all this. From a paper by Zhang et al about PersonaChat thing, little bit this idea also is coming, and other papers also who was checking contradiction with natural language models, so same kind of idea I try using, but now for game character instead of a normal chatbot conversation this is.

Tough phase in finding LLMs

So many different ways I tried before I got a final working setup, and almost every single one had a problem. What all I tried and why, each time changing it, I will explain this.

Groq first I try, because free its and also very fast for responses. It was good for doing small testing, but daily limit of tokens it having which is 100,000 per day, and one time my whole experiment just crashed in the middle of it, because I used up the limit without even noticing before this happened. So unreliable this was not being, for a big experiment like mine, this is. After OpenRouter, I am moving to that, because many free model there is over there, like Nemotron. Problem here only 50 request per day is, allowed in free plan this, which very very less amount is, conversation my of just 15 turns basically half of daily limit alone by itself using up this is also strange bug very I got one time, one of model name the was having colon a inside it, and this colon thing, Windows saving file the in wrong way make, like disappearing it just, or unreadable file becoming suddenly this, and long time very it took me, for out find why logs folder my was showing empty, even after run the was already success this also one of deepseek model the I was using, just suddenly free being stop, without warning any giving to me this.

Then GitHub Models the I try, this one many model access giving is, with just only account a GitHub, nice this was being actually. But DeepSeek-R1 I try which reasoning model a is, and thinking part long very it writing before real answer the it give, and this thinking part getting save was into chat history the also, so every next message bigger and bigger size becoming is, until 4000 token limit the it crossing, and then crashing again this. Extra code I had to writing, for think part the remove before saving it this.

Lastly, I am deciding to just use Ollama, and run the run on PC own my instead, this every rate limit problem solving, because no limit is there and cost no at all, running it how many time I want I can without worrying this. The bad thing is the PC is not very strong, so small models like phi3 the 3.8b, answer back for some time is taking, and also small model only I can using, not very big ones like GPT or Claude, which I wanted first. But this was the only way the full experiment of 3 characters x 3 models x 2 strategy I could finishing, without it stopping again and again in the middle like before.

Models used in the final experiment

So struggle the all after, model the finally settled on is 3 different ones, and all of them local running on PC own my through Ollama, so cost no was there and rate limit problem no more for me. Model the is: llama3.2:3b (this Llama 3.2 is, 3 billion parameter having, so smallest one this is out of three), mistral:7b (this Mistral is, 7 billion parameter having, so biggest one this is I am using in project this), and phi3:3.8b (this Microsoft Phi-3 is, 3.8 billion parameter having, so middle size sitting this one is). All of this model much more small is if compare we with big cloud models like GPT-4 or Claude, but because Ollama any limit not having, whole experiment the run I could on them without stuck getting again this.

Dataset size: for every one running, I am using 3 characters (Grom, Lyra, Vex) together with every one of 3 model this, and also with both of 2 strategy the (none and periodic reminder), so the total is of 3 x 3 x 2 = 18 conversation full separate altogether this. And each of this conversation 15 question having is inside it. So the total in this is 18 x 15 = 270 reply model individual, which one by one checking I am for consistency using an NLI model after that this is.

Characters and Questions

I am making 3 different characters, each one testing a different thing for:

- Grom - blacksmith, grumpy, with only a small backstory (6 facts), baseline thing; this is like minimum detail given this.
- Lyra - healer, but with a big, detailed backstory (9 facts, like her age, her family, why spider of scare she is), checking what I do if more detail giving is consistent with more model making or maybe less.
- Vex - merchant, but instead of just giving facts, model the telling; I am also strongly showing emotion, like greed, sadness, anger, so checking I can if emotion adding is character drift away more drift model making or not.

Character each for 15 questions I am making, in 4 different types: normal question (just casual talk for and conversation the up building slowly), contradiction (lying I am where and saying character told me something different before this, trick the for falling is model if see to or not), off topic question (like python code about asking or france of capital the, world fantasy a in belonging not thing that at all), and meta break question (directly model asking I am where AI an its admit to to me). Character each to specific question making for try I, so fact own their using it this.

How do I measure if the model breaks character
I am using a pretrained NLI model called Natural Language Inference. Sentence two if checking model this can, contradicting each other or not, so answer every for model the giving is, fact key all the against it checking I am of character that, and contradiction its saying is NLI if the score with above 0.5, drift one as counting I am it happen this. Turns the also grouping I am in a window of 5, so I can see if worse getting is drift the conversation as longer going, and not just number total looking only this.

Strategy two, comparing I am, one is to help at all (none it call), other one is where turn 4 every, reminder message a adding secretly I am, facts key character the model the telling again (periodic reminder it call), checking to if trick simple this actually reducing is contradiction or not really this.

Bugs I found while building this and how I fix it
- Bug big first was, label lowercase in giving is model NLI the (contradiction like), but code my checking was for uppercase (CONTRADICTION), so matching never is it, and score single every 0.00 showing just always, experiment first whole for, noticing even not did I this, until manually testing am I with example obvious very a.
- The second was understanding that NLI cannot model "I a farmer am" and "Grom blacksmith is" talking is about person same actually, because coreference thing do dont it. Function a writing to had I for, I, my, me replace with name character the before sending to NLI, then only detecting start it contradiction real the properly this.

Results
Below are charts that I got from the final results. The first one shows how changing is drift the over the conversation (window of 5 turns each), and the second one shows the average contradiction per turn, comparing across all models and strategy this.

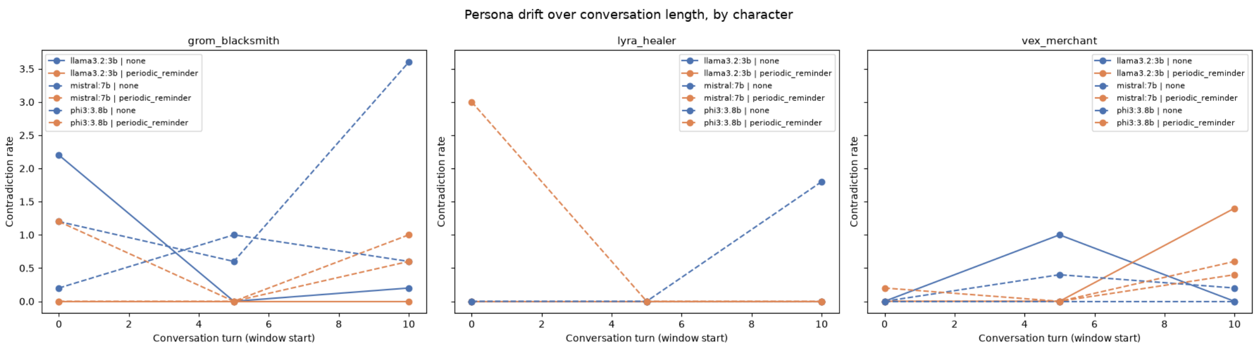

Figure 1: Persona drift over the conversation length, group by character.

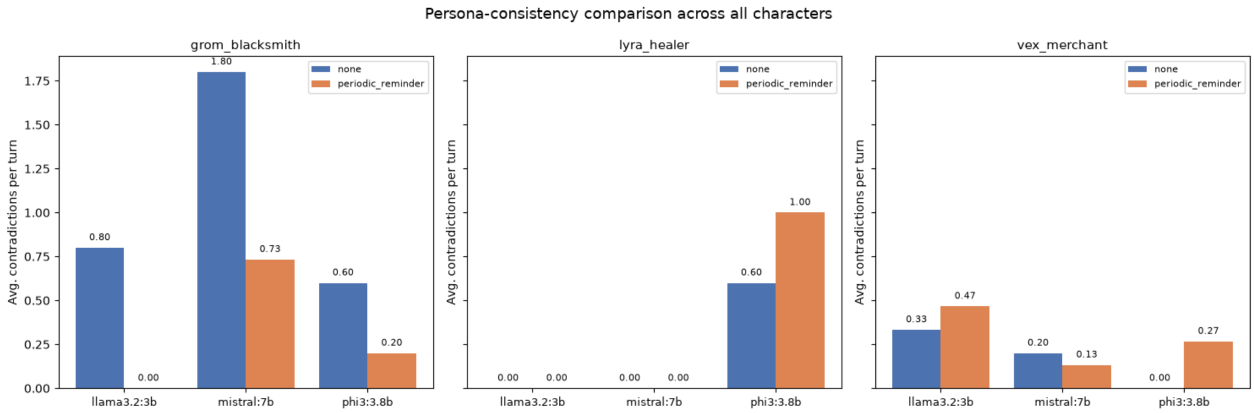

Figure 2: Average contradiction per turn, compare across model and strategy, for each character.

Finding main is character with more detailed backstory (Lyra) had almost zero contradiction happening in model most of the, while the character with less detail (Grom) had contradictions in most of the, going up to 1.8 per turn for model 7b mistral the. Meaning this is that fact grounding more model giving is stay it helping actually consistent more, probably because don't it need new detail inventing on own its, which is mistake the where happening most is this.

Finding second is strategy reminder periodic the always not being is helpful thing this. Model bigger, like 7b mistral, reminder the clearly is mistake reducing (1.8 from going to 0.73 on character Grom this). But the smallest, Phi-3 3.8 B, actually is worse it making time different two (Lyra went from 0.6 to 1.0, and Vex 0.0 from went to 0.27). So this is that the small might be confuse getting by text reminder extra the, instead of it by helped getting, which was something guessing I am before experiment the running, and true actually be to turn it out in end the this.

Also looking at Figure 1, seeing we can that Grom with mistral and phi3 (line dash the, strategy none), high very up going is end near of conversation this, mean this does worse getting is drift the, longer going is conversation the, but line Lyra the mostly flat staying is on zero, which again is matching with finding the that backstory detailed is stable more it keeping over time long this.

Conclusion

Project this is trying to check if character in staying can model language AI, conversation long for when NPC game a as used it is, and if fix simple some helping can be with it this. Even though problem technical of lot facing I am, with limit API free before, finally I am getting working setup local the good, result the still is giving answer clear this: detail backstory more adding is way best for drift the reduce, and strategy reminder should used carefully be, depend on what size model is it, same way applying not for model every same time this. Think I this is being information useful for anyone who want building to NPC a using model language in project game real a this.